

Curriculum Ordering, Catastrophic Forgetting, and OOD Generalization in PPO Agents

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Poster Presentations in context of “Reinforcement Learning” Lecture github.com/keisuke58/LUH_summer_2026

1 TL;DR

Q: Does curriculum ordering help OOD generalization?

A: **Progressive is worst**, both in-dist *and* OOD, due to catastrophic forgetting.

Diversity, not ordering, drives transfer.

Rec: Use multi-task (Mixed/Random).

3 Approach

3 training stages:

- Easy **Empty-8x8** (dense)
- Medium **FourRooms** (sparse)
- Hard **KeyCorridorS3R1** (v. sparse)

5 held-out OOD environments:

DoorKey, MultiRoom, SimpleCrossing, LavaCrossing, DistShift1

5 conditions × **10 seeds** × **1M steps**

Condition	Rule
Progressive	Easy→Mid→Hard ($\tau=70\%$)
Reverse	Hard→Mid→Easy
Random	Uniform random
Hard-only	Always Hard
Mixed	Multi-task

Algorithm: PPO (stable-baselines3)

Advance rule: success rate $\geq \tau$ over 20-episode window

Stats: Kruskal–Wallis + Mann–Whitney U (Bonferroni $\alpha=0.05$)

5 Future Works

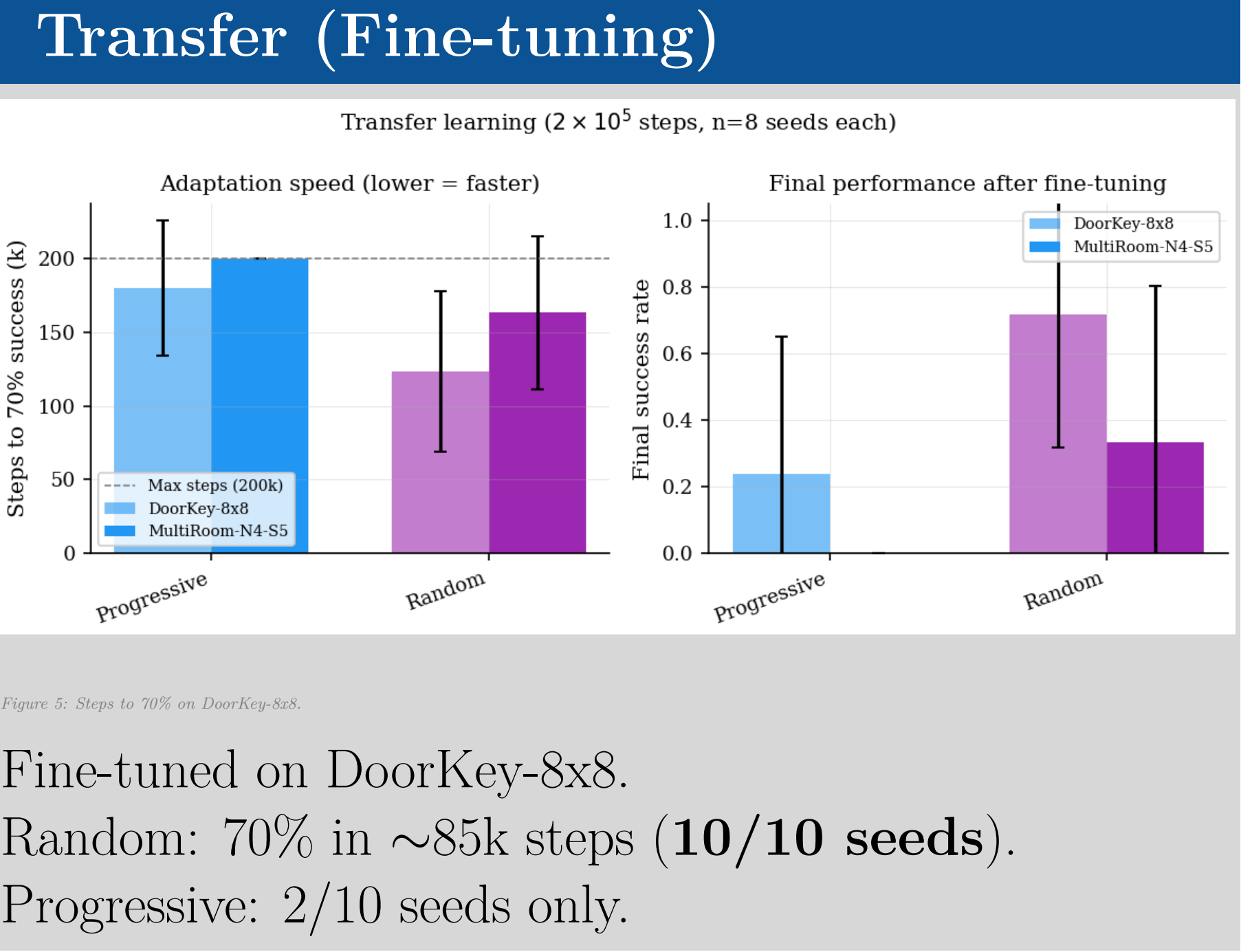
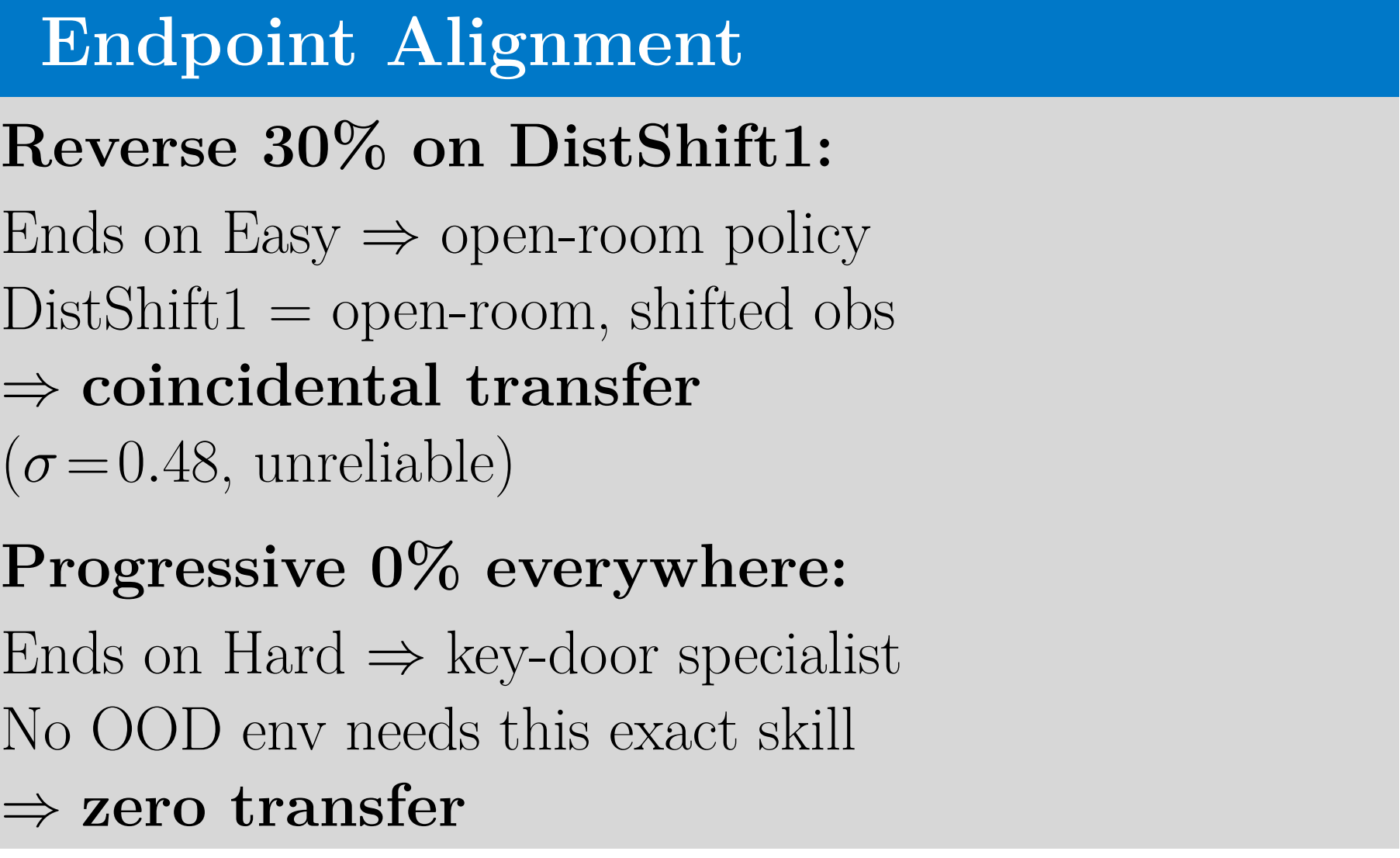
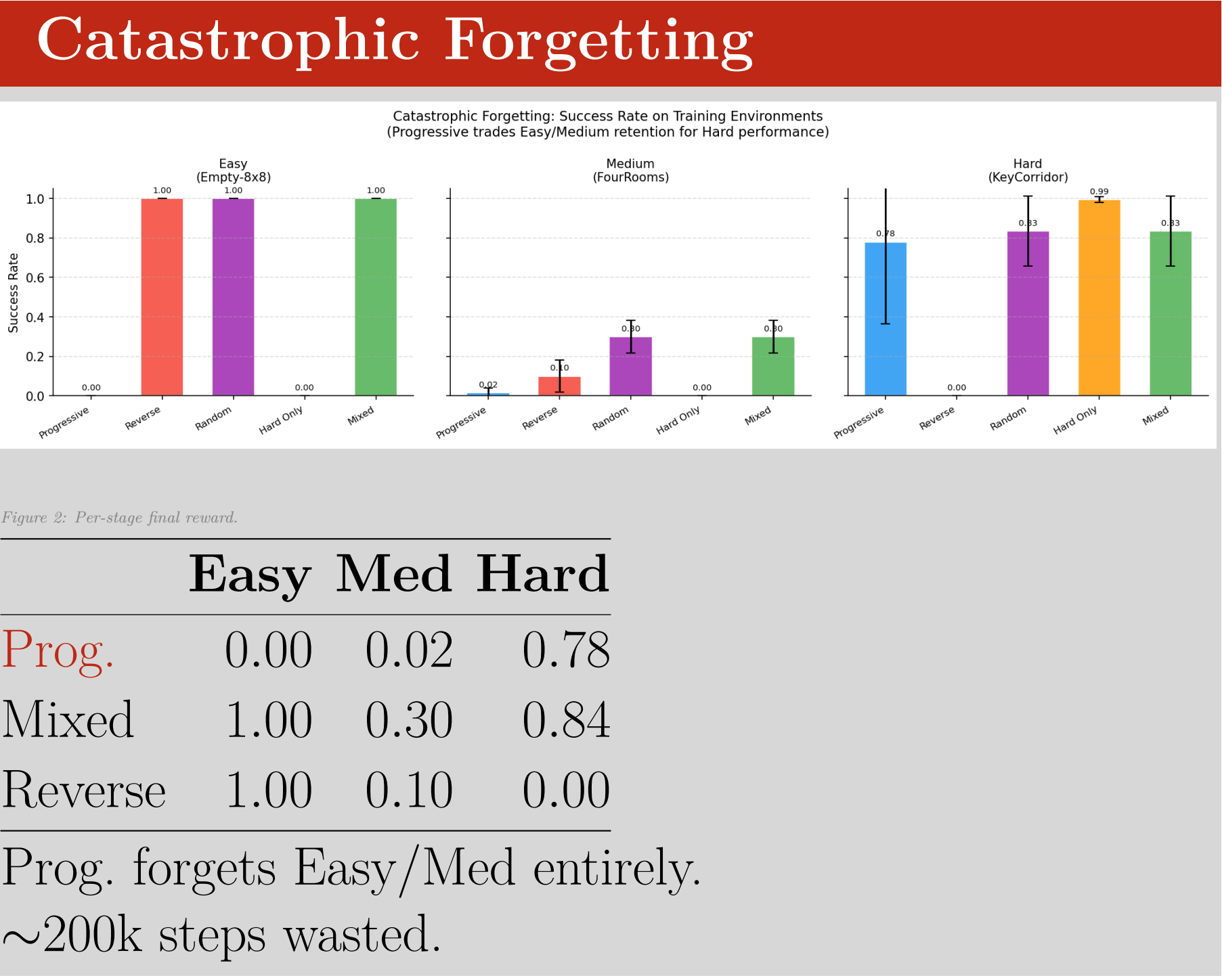
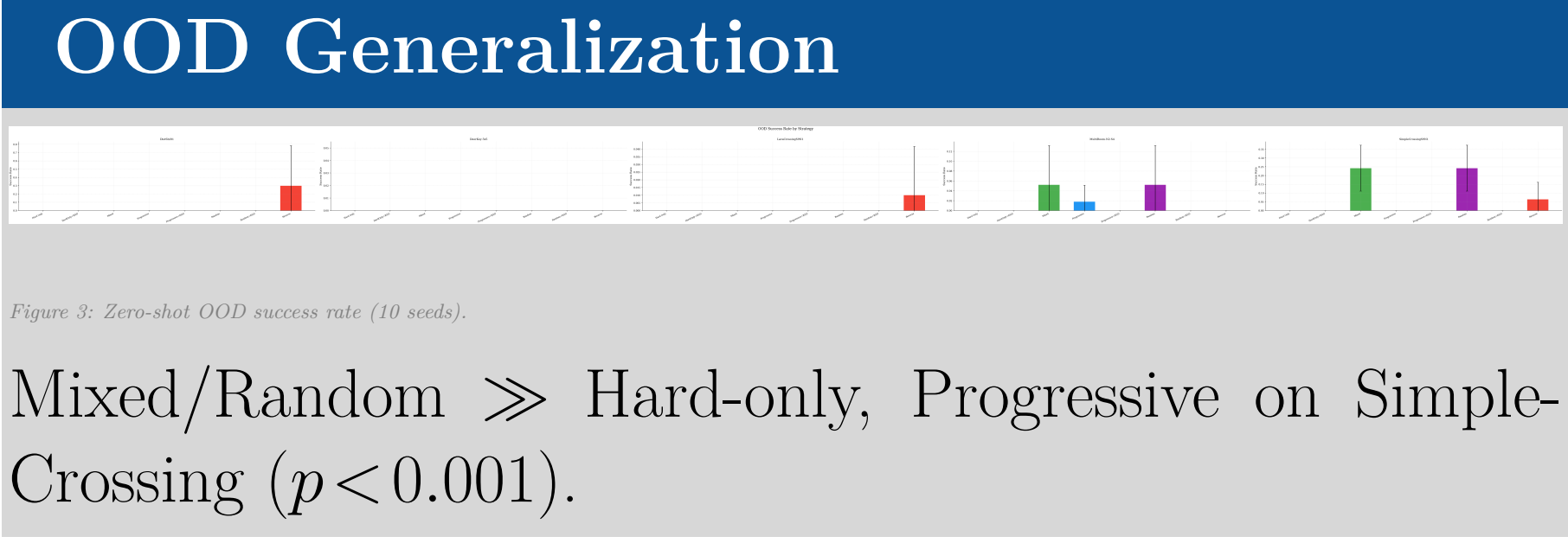
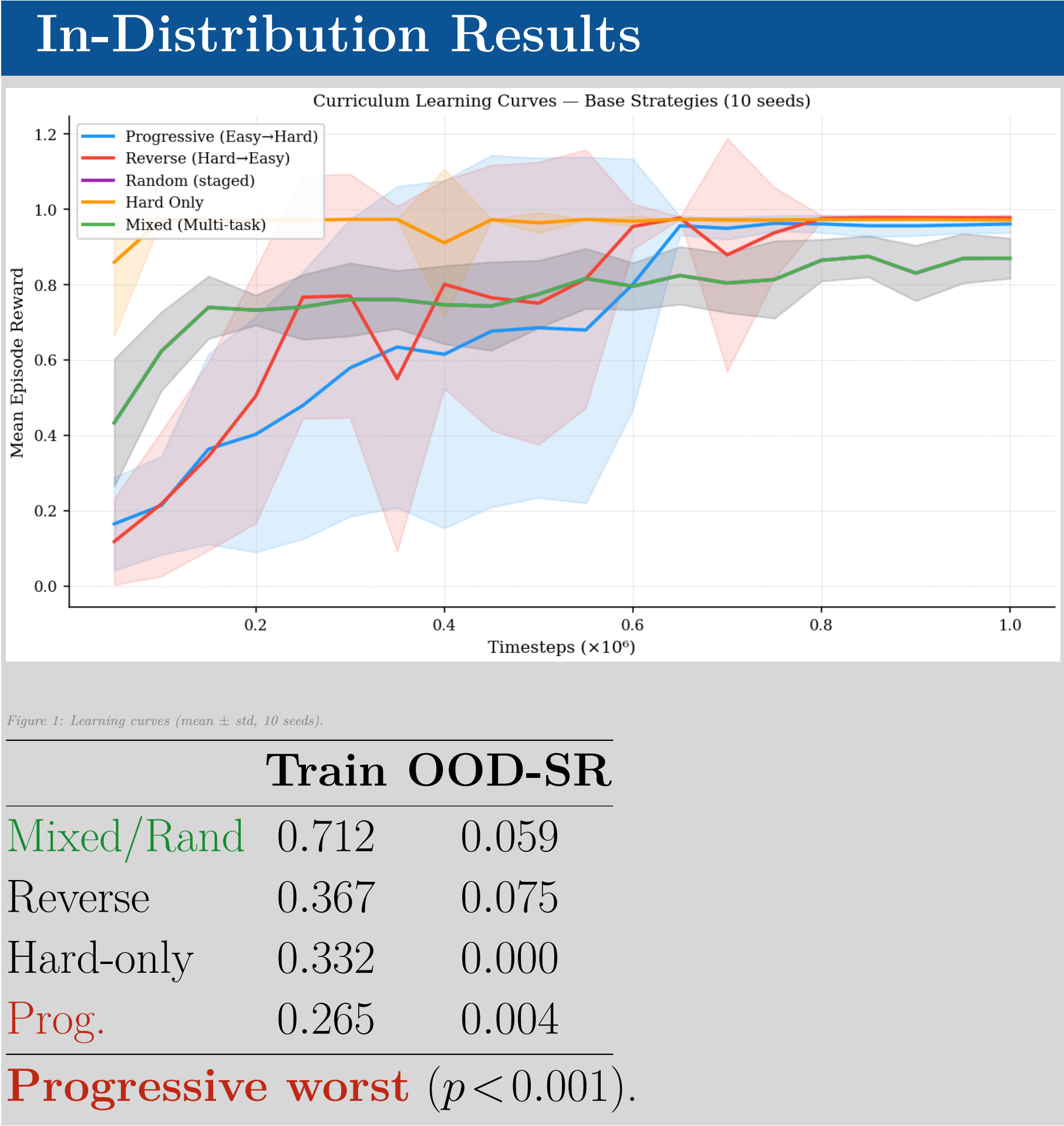
- Auto-curriculum** [Portelas et al., 2020]: dynamic difficulty avoids forgetting
- EWC** [Kirkpatrick et al., 2017]: regularize stage-transition forgetting
- Domain randomization** for principled OOD transfer
- Scale to 10M steps, more diverse envs

2 Motivation & Problem Setting

Curriculum learning [Bengio et al., 2009] proposes easy-to-hard ordering to accelerate convergence, but whether ordering also promotes **OOD generalization** [Kirk et al., 2023] is unknown.

Hypothesis	Expected
H1 Progressive > oth-ers in-dist	Easy→Hard eases learning
H2 Mixed best for OOD	Diversity promotes transfer
H3 Perf.-gen. gap	Curriculum overfits training

4 Key Insights



Key Takeaway: Environment *diversity*, not curriculum ordering, drives performance and OOD generalization. Use multi-task training.